Problem B. LRUD Moving

Time Limit 2000 ms

Problem Statement

You are given positive integers N, A, B. It is guaranteed that both A and B are between 1 and N, inclusive.

There is an $N \times N$ grid. The cell at the i-th row from the top and j-th column from the left is denoted as cell (i, j). Initially, a piece is placed at cell $(1, 1)$.

By repeating $N^2 - 2$ moves, each of which moves the piece to an adjacent cell (up, down, left, or right), you want to move the piece to cell (N, N) while visiting every cell except cell (A, B). You must not visit the same cell more than once (you must not visit cells $(1, 1)$ and (N, N) in the middle, either).

Determine whether such a sequence of moves is possible, and if so, output one such sequence.

You are given T test cases; solve each of them.

Constraints

- $1 \leq T \leq 5000$
- $2 \leq N \leq 10^3$
- $1 \leq A, B \leq N$
- $(A, B) \neq (1, 1), (N, N)$
- The sum of N^2 over all test cases is at most 10^6.
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$\begin{array}{l} T \\ \text{case}_1 \\ \text{case}_2 \\ \vdots \\ \text{case}_T \end{array}$$

Each test case is given in the following format:

$$N \ A \ B$$

Output

Output the answers for the test cases in order, separated by newlines.

For each test case, if it is impossible to perform a sequence of moves satisfying the conditions, output `No`.

If it is possible, output in the following format:

```
Yes
S_1S_2...S_{N^2-2}
```

Here, S_k is defined as follows, where the coordinates of the piece before the k-th move is cell (i, j).

- $S_k =$ `L` if the piece moves from cell (i, j) to cell $(i, j-1)$
- $S_k =$ `R` if the piece moves from cell (i, j) to cell $(i, j+1)$
- $S_k =$ `U` if the piece moves from cell (i, j) to cell $(i-1, j)$
- $S_k =$ `D` if the piece moves from cell (i, j) to cell $(i+1, j)$

If multiple sequences of moves satisfy the conditions, any of them will be accepted.

Sample 1

Input	Output
3 2 1 2 3 2 2 4 3 2	Yes DR No Yes RRRDLLLDDRRURD

Consider the first test case.

Starting with the piece at cell $(1, 1)$, move it twice as follows:

- Move the piece downward. The piece moves to cell $(2, 1)$.
- Move the piece rightward. The piece moves to cell $(2, 2)$.

This sequence of moves satisfies the conditions.